

Oatmeal on printers

A hilarious take on printers from online web comic giant Oatmeal : <http://theoatmeal.com/comics/printers>

PSD, SVG on Google Docs

Google Docs has added support for 12 more formats to its list including .PSD, .PPTX, .XLSX among others

quality in all the test regions giving crisp colours, great skin tones and detailing in complicated patterns. Brother MFD took around seven minutes to print the page and we noticed a warm overall cast in the output. This made the skin tones look good, but it didn't really earn many points in the other regions of the photo.

Copying test was good across all the MFDs with just HP 8500A giving a noisy output. There was no smudging. Again, Brother MFD took the longest with 91 seconds to copy.

Scanning quality was also quite similar across all the MFDs. HP 8500A took a mere 13 seconds to scan the magazine cover at 300dpi. MFDs like HP 8500A, C310a, Brother J615W and Epson 960FWD offered scanning to email, USB drive and memory card (wherever applicable).

Setting up Wi-Fi on the MFDs supporting it was quite easy. Each one comes with a detailed manual to guide you through the steps. Epson's manual, was the most detailed with proper flowcharts explaining the Wi-Fi setup process. Web connected MFDs from HP were quite impressive but the resistive touchscreen is not as fluid as we would have liked. Also we would prefer if HP enabled more applications for the Indian

market, as the HP ePrint centre has a lot of apps on offer. We did face some issues with the ePrint on both the MFDs.

Verdict

When it came to the final verdict, we had to take into consideration not just the print quality but also the time taken to print. Features had an important weightage, Quality of ADF scanning and scanning to USB drives and memory



Canon PIXMA MP497 gave good quality prints, but has expensive cartridges

was not scored as some MFDs did not provide that function, but the feature was scored. Total cost of ownership was taken into consideration with Best Buy calculation.

According to our tests, **HP OfficeJet**

Pro 8500A Plus was the awarded the **Best Performer**. Not only did it give good quality prints, but did the job fast. HP 8500A quality was slightly lower than that of HP C310a and was at par with

Canon's MP 287, but the speed is where HP 8500A scored higher than the others.

Epson ME Office 960FWD was awarded the **Best Buy** as it gave the best price to performance ratio. Although the overall score of the 960FWD was higher than that of HP 8500A, in terms of performance only, the 8500A was better than 960FWD. Despite scoring highest on the total cost of ownership, Epson 620F lost out to 960FWD on the feature set. For the same cartridge cost, you get better features on 960FWD at a higher price and better performance. Canon lost out on the Best Buy, despite having good quality prints and lower MFD price, due to the overpriced ink cartridges. **d**

HOW WE TESTED

All the multifunctional devices that we got were tested on our standard printer test bench – Intel Core i7 965 CPU, 2x2GB of Corsair DDR3 RAM. This year, we decided to concentrate only on colour MFD's for home and SoHo use. We also kept the maximum price point around ₹25,000, because we don't think anyone in a home or SoHo setup will spend more than this on an MFD.

As expected, all the MFDs were inkjet-based. We ran a few dry runs with the MFDs before actually starting off with our tests. We used standard A4 size paper for normal print and copy tests, and Kodak Photo glossy paper for photo prints.

The print tests were standard printer tests – quality of text printing, mixed document printing and photo printing. The text-only documents were reviewed for clarity of the smallest and largest fonts, and we looked for evenness of the text at different settings. In the mixed document tests we tested the prints against each other for clarity of text, merging colours, changing gradations of colours and white to black progressions. In our photo test, we printed an A4 size test photo, and checked for the sharpness levels, colour output and also looked for flaws such as merging of colours or grain.

For the mixed document and photo prints, we identified over five regions where printers usually struggle, and scored the outputs out of 10. The final quality score was an average of all the individual test scores.

For the copy tests, we compared the output to the source and checked for noise, change in colour level, clarity, etc. Each test was timed, and that was also scored. Scanning speeds and quality were also tested in all the MFDs.

Features of MFDs often affect buying decisions, and we gave considerable importance to this. We tested the ease with which cartridges could be replaced, how easy was the MFD to setup, and of course how user friendly was the software interface. We also felt that build quality of MFDs is important, and scored it accordingly. We did not really test the fax functionality, as some MFDs lacked it, and who uses fax any more?

What's new this year is the way we calculated total cost of ownership (TCO). We did this to ensure that those MFDs which were priced low, but had high running costs weren't given any undue advantage. We assumed that the life of an MFD in SoHo conditions is 15,000 prints, and calculated the cost of owning the MFD for that long – this included calculations with cost of the device, cost of cartridges, and thus cost per print, and also the warranty period and the assumed maintenance costs. The TCO index we got was considered in the overall score of the MFD.

Brand	Contact	Phone	Email	Web site
Brother	Brother International India Pvt. Ltd.	1800-222 422	info@brother.in	www.brother.in
Canon	Canon India Private Limited	1800-345 3366	info@canon.co.in	www.canon.co.in
Epson	Epson India	+91-80-3051 5000	think@eid.epson.co.in	www.epson.co.in
HP	HP India	+91-124-256 6111	bharatkharbando@hp.com	www.hp.com